Mishkal Tahor - User Manual

Version V14 (beta) · Mishkal by Lovy Elias · Nov 7, 2026

Welcome

Thank you for testing Mishkal Tahor. Tahor is three EQs in one window: **Reso** tames harsh resonances, **Tahor** is a clean 8-band surgical EQ, and **Character** adds the colour of vintage-style hardware EQs. It is part of the Mishkal plugin collection by Lovy Elias.

This manual covers version **V14 (beta)**. Tahor runs as VST3, AU and AAX on Mac, and VST3 on Windows.

The sound always flows in this order:

Input > Reso > Tahor > Character > Output

So Reso cleans the sound first, Tahor shapes it, and Character colours the result.

Quick start

1. **Insert Tahor** on a track (a vocal is a great first test) and press play. The analyzer only appears while audio is playing.
2. **Set your level.** Click **AUTO** next to IN in the bottom bar. Tahor listens for about 3 seconds and sets the input to sit around -18 dBFS RMS.
3. **Pick a starting point.** Click the star next to the preset name to open the preset browser, and choose a preset from a folder such as Vocals or Drums.
4. **Choose a section.** Use the RESO / TAHOR / CHARACTER selector at the top (or press 1, 2 or 3). Double-click the screen in Reso or Tahor to add a band, then drag it.
5. **Compare fairly.** Turn on **MATCH** and use **BYPASS** or **A > B** to hear whether your changes really make it better, not just louder.

The window at a glance

The window has four parts: the header along the top, the big screen in the middle, a meter on each side, and the bottom bar. Everything here works the same in all three sections.

Header (top row, left to right)

Control	What it does
Tahor name	Shows which plugin you are in.
Undo / Redo	Steps back or forward through your changes.
History	Opens a list of your recent changes by name, so you can jump back several steps at once.
< > beside the preset name	Steps through presets one at a time.
Star	Opens the preset browser.
Preset name	Click it to open the preset menu. Presets are organised into folder submenus. The menu also has Save, Save As (with a tag) and Browse presets.
A > B	Compares two versions of your settings.
MATCH	Matches output loudness to the input, so comparisons are fair.
BYPASS	Switches the whole plugin off so you can hear the original.
RESO / TAHOR / CHARACTER selector	Chooses the section you are editing. The arrows either side flip through the three. The power icon switches the current section on or off; a switched-off section goes dull.
AUTO GAIN	Estimates how much louder or quieter your EQ makes the sound and corrects it at the output. Shows the correction in dB.
Mishkal logo	The Mishkal collection badge.

The screen

The large middle area shows your EQ curves over a live analyzer. The darker layer is the sound **before** Tahor (pre) and the brighter layer is the sound **after** it (post). The analyzer only appears while audio is playing.

The frequency scale along the bottom runs from 20 Hz to 20 kHz, with both ends labelled. The square button in the top-right corner of the screen switches to full screen.

Band panels

When you touch a dot in any section, its band panel opens at the bottom of the screen. All three sections share the same modern look: smooth dark glass that is 80% solid (you can see a little through it), softly angled corners, a small arch rising toward its band, and a thin line in the band's colour along the top.

- **Power button (top-left):** switches the band off but keeps its settings. The dot stays on screen, dimmed.
- **Big main knobs:** the most-used controls are larger; secondary ones are smaller.
- **Placement dropdown:** chooses where the band works (for example Stereo, Left or Right). Choosing Left or Right makes an independent partner band on the other side.
- **S:** solo the band.

The bubble

The bubble above a dot you touch shows its values live, and has a small **S** button to solo that band. This works in all three sections.

Every part of the bubble can be changed right there, like in Pro-Q:

- **Shape:** click the shape name (Bell, Low shelf, High shelf, Low cut, High cut, Notch) for a menu.
- **Frequency, gain and Q:** click the value and type a new one, then press Return (Esc cancels). Frequency takes "2500", "2.5k" or a note name such as "A4" or "C#3"; gain takes "-3"; Q takes "0.7".
- **Stays open:** the bubble stays open while your mouse is over it and lingers about 4 seconds after you leave, so the S (solo) button is easy to reach.

Cursor readout: move the mouse over an empty part of the RESO or TAHOR screen and a thin line follows it, with a readout above the frequency scale showing the frequency and note under the mouse (on TAHOR, the gain at that height too), for example "3.5 kHz · A7 +7c".

Right-click menu

Right-click (or two-finger click) any dot for a menu of quick actions. Each section's menu is listed in its own chapter.

Meters

The input meter runs down the left edge and the output meter down the right edge. At the bottom of each you will see two readouts: **PK** (the peak, held) and **AVG** (the average level).

Bottom bar (left to right)

Control	What it does
IN	Input gain.
AUTO	Listens for about 3 seconds and sets the input so it sits around -18 dBFS RMS.
Phase mode	**Natural Phase** (low latency), **Linear Phase**, or **Linear Phase HQ**.
Oversampling	Off up to 32x. Mainly helps the Character section's colour sound smooth.
Render	The oversampling used when you bounce or export.
Keyboard button	Switches the piano roll (note names along the bottom of the Tahor screen) on or off. Off by default.
Analyzer	Pre + Post, Post, Pre or Off.
Range	The display scale of the screen: ±3, ±6, ±12 or ±30 dB. It changes the view, not the sound.
Size	Window size, from Mini to Extra Large. Tahor opens at Small (1064 x 680) the first time.
SC Ext	Uses the external sidechain input from your DAW.
OUT	Output gain.

Reso: the resonance suppressor

Reso finds harsh, ringing frequencies and dips them only at the moments they ring. The rest of the time it leaves the sound alone. Use it for harsh vocals, boxy rooms, fizzy guitars and sibilance.



Figure 1. The Reso section, with the Depth panel on the left and a band panel open.

Reading the screen

- **The red curve at the top** shows what Reso is removing, in real time. It hangs down from the top of the screen; the deeper a dip, the more Reso is cutting there.
- **The red edges** (vertical lines at the left and right) are the focus range. Drag them in to make Reso work only between those frequencies.
- **The dots** are shaping bands. They tell Reso to push harder (more) or relax (less) at a frequency. They do not boost or cut the sound directly.

Shaping bands

Double-click an empty part of the screen to add a shaping band; double-click a dot to remove it. Shapes available: Bell, Bandpass, Low shelf, High shelf, Tilt, Notch, Low cut and High cut.

Touch a dot to open its band panel:

- **Power button (top-left):** switches the band off but keeps its settings; the dot stays on screen, dimmed.
- **Amount** (big knob): how much more (+) or less (-) Reso works at that frequency.
- **Width** (smaller knob): how wide the band is (shown as Q).
- **Placement dropdown:** Stereo, Left or Right. Choosing Left or Right makes an independent partner band on the other side.
- **S:** solo, to hear just that area.
- **Shape:** the dropdown at the bottom of the panel.

The bubble above the dot shows its number, shape, frequency and amount, for example "2 · Bell · 970 Hz · +3 dB · Q 1.20", with a small **S** button to solo it.

Right-click menu (or two-finger click on a dot): Enable/Disable, Invert amount, Solo, Shape, Stereo placement, Cut, Copy, Paste, Split (left/right), Delete.

Left panel

The main Reso controls live in the panel on the left. Click the small tab on its edge to slide it in or out. The panel is 60% see-through, so you can still see the analyzer behind it.

Control	What it does
Soft / Hard	Gentle, smooth suppression or a firmer, more obvious one.
Low latency	Reduces delay, useful while recording or playing live.
Depth	How strongly Reso cuts overall. Start here.
Detail	How narrow and precise the cuts are. Higher values catch smaller, sharper resonances.
Attack	How quickly Reso reacts when a resonance appears.

Control	What it does
Release	How quickly Reso lets go after it stops.
Max cut	The most Reso is ever allowed to cut, for example -12 dB.
Mix	Blends the processed and original sound.
Trim	Output level of the Reso section.
Reduction	A meter showing how much Reso is cutting right now.
Stereo / Mid / Side	Process the whole signal, only the middle, or only the sides.
Delta	Lets you hear only what Reso is removing. Great for checking you are not cutting too much.
Sidechain	Lets another signal decide when Reso works.

Advanced panel

Click the **ADV** tab on the right edge to open the Advanced panel. Like the left panel, it is 60% see-through.



Figure 2. Reso with the Advanced panel open on the right.

Control	What it does
Detail tilt	Leans Detail toward the lows or the highs.
Attack tilt	Leans Attack toward the lows or the highs.
Release tilt	Leans Release toward the lows or the highs.
Stereo link (Link)	How much the left and right channels react together. 100% means they move as one.
Phase: Minimum / Linear	Linear keeps parallel and Mid/Side processing working cleanly.

Tip: turn on Delta and raise Depth until you hear only the harsh ringing in the Delta signal, then turn Delta off.

Tahor: the surgical EQ

Tahor is a clean, precise EQ with up to 8 bands. Use it to remove mud, notch out a ring, cut rumble or gently lift a frequency.

Working with bands

- **Add a band:** double-click an empty part of the screen.
- **Remove a band:** double-click its dot (or press Delete).
- **Move a band:** drag its dot left or right to change the frequency, up or down to change the gain.
- **Change the width (Q):** roll the mouse wheel over the dot.
- **Read the values:** the bubble above the dot you touch shows its shape, frequency with the note name, and gain, for example "Bell · 1 kHz (B5 +21c) · +4 dB · Q 1.00". Click its small **S** to solo the band.

The band panel

Touch a dot to open its panel at the bottom of the screen.

- **Power button (top-left):** switches the band off but keeps its settings; the dot stays on screen, dimmed.
- **Shape dropdown:** Bell, Low shelf, High shelf, Low cut, High cut or Notch. The cuts have slopes from 6 to 48 dB per octave.
- **Placement dropdown:** Stereo, Mid, Side, Left or Right.
- **Knobs:** Frequency and Gain are big; Q is smaller.
- **OPTIONS dropdown:** chooses which controls the panel shows (**EQ controls** or **Dynamic controls**), and has **Dynamic on** and **Sidechain**.
- **S:** solo the band.

Right-click menu (or two-finger click on a dot): Enable/Disable, Invert gain, Make dynamic/static, Solo, Shape, Slope (cuts), Stereo placement, Cut, Copy, Paste, Split (left/right), Delete.

Dynamic bands

A dynamic band only moves when the sound crosses its threshold, so it cuts a harsh note only while it plays. Turn it on with **Dynamic on** in the OPTIONS dropdown (or **Make dynamic** in the right-click menu). Dynamic stays active when the panel shows the EQ

controls.

Choose **Dynamic controls** in OPTIONS to see:

Control	What it does
Threshold (big)	The level where the band starts to act.
Range (big)	The most the band can move, for example -8 dB.
Attack	How fast it reacts.
Release	How fast it lets go.
Q	The width of the band.
Live meter	How many dB the band is pulling right now.
Power button (next to the meter)	Turns Dynamic on or off.

On screen:

- **Solid curve:** the gain you set.
- **Shaded curve:** the reduction happening right now.
- **Dashed outline (range):** the full range the band may move. Grab it anywhere along the outline and drag up or down to set the range directly; the outline brightens when your mouse is over it.
- **Diamond:** a small diamond on the range outline marks the band's centre.



Figure 3. A dynamic band in Tahor, panel showing the EQ controls: set-gain curve, shaded reduction, dashed range outline with the diamond at the band's centre, and the bubble with the note name.

Splitting a band into left and right

Choose **Left** or **Right** in the placement dropdown (or **Split** in the right-click menu). Tahor makes a partner band on the other side with the same settings. From then on, each is independent, so you can set the left and right differently.

- The left curve is **blue** and the right curve is **pink**.
- The dots are tagged **L** and **R**.
- Choose **Stereo** to join them again.

Solo

Press **S** in the band panel or the bubble (or the S key). The soloed band's range lights up as a column on the screen, as wide as its Q, and the rest of the screen dims. You hear only that part of the sound, which makes it easy to find a problem frequency.

Spectrum Grab

Hover near a peak in the analyzer and a "grab" marker appears. Drag it and it becomes a narrow band that you are holding, ready to pull the peak down.

Selecting several bands

Drag a box across the empty background, or Shift-click dots, to select several bands. Drag any selected dot to move them all together. Press Delete to remove them all.

Piano roll

Click the small keyboard button at the start of the right side of the bottom bar to show note names along the bottom of the Tahor screen. It is off by default.



Figure 4. A band split into left (blue) and right (pink), a soloed band's lit column, and the piano roll switched on.

EQ Match

EQ Match makes your sound's tone match a reference track.

1. Route the reference track into Tahor's sidechain.

2. Click **EQ MATCH** in Tahor's top row and play both for a few seconds. The button shows LEARNING and counts the seconds.
3. Click it again to **APPLY**. Tahor places up to six bells so your sound's tone matches the reference. Level differences are ignored.

Character: vintage-style colour

Character adds the tone of classic hardware EQs. You can chain up to 4 units, each inspired by a famous piece of studio gear and given its own LE name.

Adding units

Character starts empty: nothing is in the chain. LE-EP1 is shown only as a preview; it is not added until you add it.

1. Click **+ Add EQ** to open the gear gallery.
2. Click a faceplate to add that unit.
3. Repeat for up to 4 units in a chain.



Figure 5. The + Add EQ gear gallery, showing all 16 units with their one-word hints.

The units in your chain appear as chips on the left of the screen, in the order the sound passes through them. Drag a chip to reorder, or click its **x** to remove it.

The faceplate

The hardware faceplate at the bottom of the screen shows the selected unit. Hover over it and arrows appear either side, so you can preview other units. A one-word hint (such as "Weight") sits quietly above the gear and tells you what the unit is best at.

The larger two-rack-unit gear (LE-EP1, LE-MP2, LE-Esh, LE-Even and LE-Kavod) is laid out like the originals: big gain knobs on top, frequency selectors under them. LE-EP1 follows the Pultec panel: **BOOST / ATTEN / BANDWIDTH / BOOST / ATTEN** over **LOW FREQ CPS / HIGH FREQ KCS / ATTEN SEL KCS**. It now has high attenuation too.

Reading the curves

- Each unit draws its **own coloured curve**.
- The **bright line** is the combined sound of the whole chain.
- The legend at the top left shows which colour belongs to which unit.

Nodes

The numbered dots on the curve drag just like Tahor's bands, but they snap to the steps the hardware offers. When a unit boosts and cuts the same frequency (the Pultec-style trick), the two dots sit slightly apart, joined by a line.

- **Double-click a dot** to switch that band off or on.
- **Alt-click a dot** to reset it.
- The bubble above a touched dot has a small **S** button to solo it.

Touch a dot to open its node panel:

- **Power button (top-left):** switches the band off but keeps its settings; the dot stays on screen, dimmed.
- **Knobs:** Freq and Gain are big; Width is smaller.
- **Placement dropdown:** Stereo, Left or Right. Choosing Left or Right splits the unit into separate left and right settings.
- **S:** solo the node's range.

Right-click menu (or two-finger click on a dot): Enable/Disable, Invert gain, Solo, Stereo placement, Reset.

Colour and THD

The top row has the colour switch, **Clean / Subtle / Warm**, and the **THD** slider, which sets how much harmonic colour is added. 50% is the normal amount. For the smoothest colour, raise **Oversampling** in the bottom bar.



Figure 6. Character with LE-EP1 and LE-Shemen in the chain and the LE-EP1 faceplate on screen.

The 16 units

All units are inspired by classic hardware. They are Mishkal's own designs, not the original brands.

Unit	Inspired by	Hint
LE-EP1	Pultec EQP-1A-style tube program EQ	Weight
LE-ME5	Pultec MEQ-5-style midrange EQ	Mids
LE-E4K	British E-series console EQ	Bite
LE-N73	1073-style preamp EQ	Warmth

Unit	Inspired by	Hint
LE-N81	Class A 81-style EQ	Girth
LE-550B	550B-style stepped EQ	Punch
LE-K9	9000K-style console EQ	Clean
LE-H69	Helios 69-style EQ	Grit
LE-SP1	Air EQ	Air
LE-M4	EQ4-style broad bands with Air	Open
LE-MP2	Massive Passive-style passive tube EQ	Gloss
LE-Esh	HG-Q-style tube EQ	Fire
LE-Shemen	Sontec MES-432C-style mastering parametric	Silk
LE-Kav	SPL PQ-style precision mastering EQ	Precision
LE-Even	Passeq-style passive EQ	Foundation
LE-Kavod	EAR 825Q-style passive tube mastering EQ	Elegance

Presets

Tahor comes with 69 factory presets, sorted into folders. They are starting points: load one, then adjust it to your sound.

Browsing

- Click the **star** next to the preset name to open the browser. Click a folder chip at the top (All, Vocals, Drums and so on) to filter the list, then click a preset to load it.
- Click the **preset name** to open the preset menu, where presets are organised into folder submenus.
- Use the **< >** arrows beside the preset name to step through presets one at a time.
- Press **Esc** to close the browser.

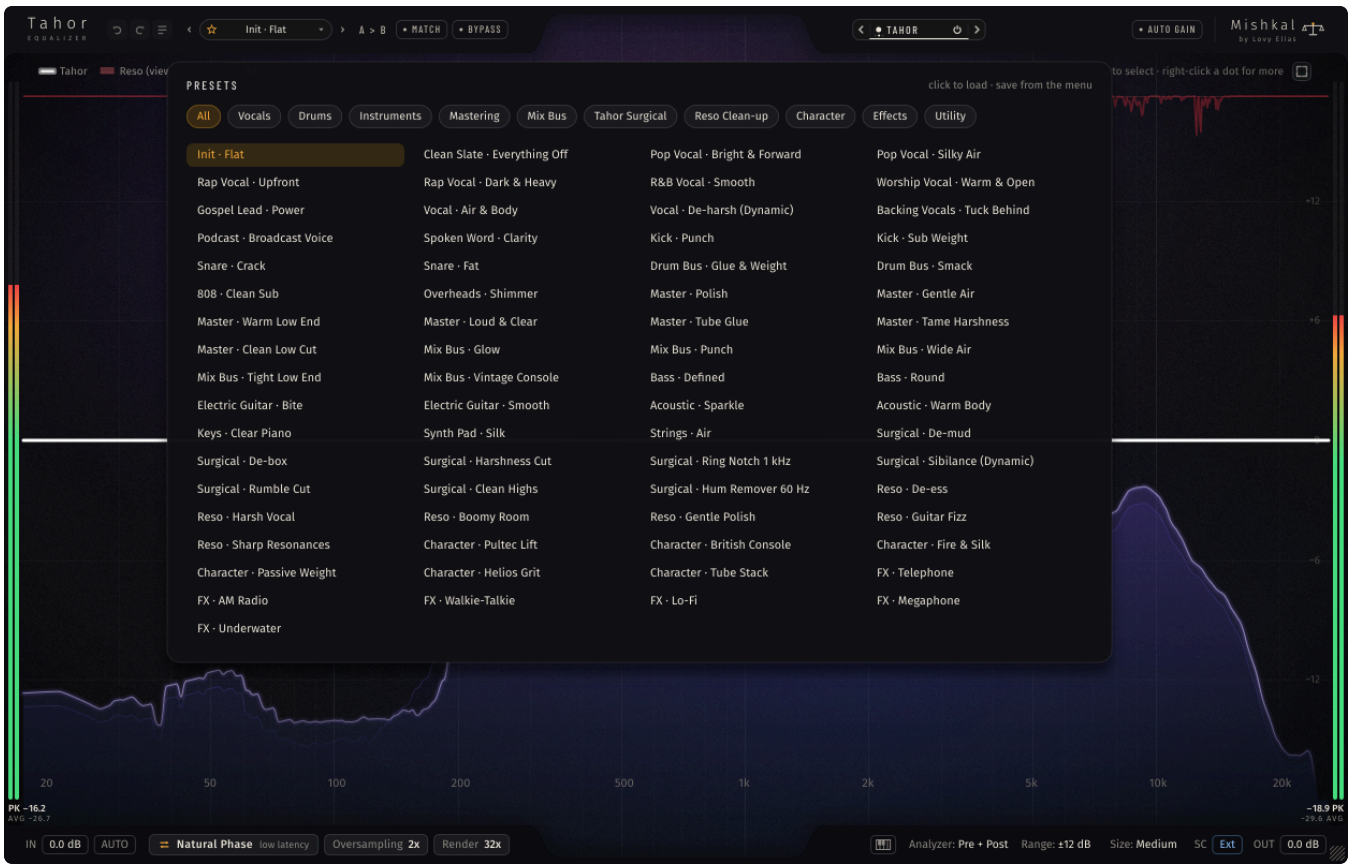


Figure 7. The preset browser with its folder chips.

Factory folders

Folder	What's inside
Vocals	Pop, rap, R&B, worship, gospel, podcast, spoken word, backing vocals
Drums	Kick, snare, drum bus, 808, overheads
Instruments	Bass, guitar, acoustic, keys, pads, strings
Mastering	Polish, air, low end, loudness and tone for the master bus
Mix Bus	Glow, punch, width and low-end control for the mix bus
Tahor Surgical	De-mud, de-box, harshness, ring notch, sibilance, rumble, hum remover
Reso Clean-up	De-ess, harsh vocal, boomy room and more

Folder	What's inside
Character	Colour chains built from the vintage-style units
Effects	Telephone, AM radio, walkie-talkie, lo-fi, megaphone, underwater
Utility	Init (flat start) and Clean Slate (everything off)

Saving your own

Click the preset name to open the preset menu:

- **Save** updates the current preset.
- **Save As** saves a new preset with a tag. The tag becomes its folder in the browser. If you don't choose one, it goes under **Mine**.
- **Browse presets** opens the browser.

Your presets are stored on Mac in:

`~/Library/Audio/Presets/Mishkal Audio/Tahor`

Metering and gain

Louder almost always sounds "better", so Tahor gives you three ways to keep levels honest.

- **IN AUTO** (bottom bar): listens for about 3 seconds and sets the input gain so the sound sits around -18 dBFS RMS.
- **AUTO GAIN** (header): estimates how much louder or quieter your EQ settings make the sound, and corrects it at the output. The correction is shown in dB next to the button.
- **MATCH** (header): matches the output loudness to the input, so when you compare with BYPASS or A > B you hear the change in tone, not a change in volume.

Meters and readouts

- **Input meter:** the left edge of the window.
- **Output meter:** the right edge of the window.
- **PK:** the highest peak, held so you can read it.
- **AVG:** the average level, which is closer to how loud it feels.

Compare the input and output readouts to see how much your settings change the level. Use **OUT** in the bottom bar for the final level.

Tip: in Reso, the Reduction meter in the left panel shows how much is being cut. In a Dynamic band in Tahor, the small live meter shows how many dB the band is pulling.

Keyboard and mouse shortcuts

Shortcut	What it does
1 / 2 / 3	Switch to Reso / Tahor / Character
Cmd-Z	Undo
Shift-Cmd-Z	Redo
Delete or Backspace	Remove the selected band (Tahor or Reso); with several selected, removes them all
Left / Right arrow	Nudge the selected band's frequency
Up / Down arrow	Nudge the selected band's gain
Shift + arrow	Nudge in finer steps
S	Solo the selected band
B	Bypass
M	Match
Esc	Close the gear gallery or the preset browser
Double-click empty screen	Add a band (Tahor or Reso)
Double-click a dot	Remove it (Tahor or Reso); switch the band off/on (Character)
Alt-click a knob or dot	Reset it
Shift-click a dot	Add it to the selection
Drag across empty background	Select several bands (Tahor)
Right-click (or two-finger click) a dot	Open its menu
Mouse wheel over a dot	Change its Q (width)

Installing

Requirements

	Mac	Windows
System	macOS 12 or later	Windows 10 or 11, 64-bit
Processor	Apple Silicon or Intel	64-bit
Formats	VST3, AU, AAX	VST3

Mac

1. Run the installer **.pkg**.
2. This beta is not signed yet, so macOS may say it can't be opened. If it does, go to **System Settings > Privacy & Security** and click **Open Anyway**. (Or right-click the installer and choose **Open**.)
3. Restart your DAW, or rescan your plug-ins.

Windows

1. Run the installer **.exe**.
2. If SmartScreen shows a warning, click **More info**, then **Run anyway**.
3. The VST3 is installed to `C:\Program Files\Common Files\VST3`. Restart your DAW, or rescan your plug-ins.

Feedback

This is a beta, and your ears help shape the final version. Please tell Lovy:

- What you used Tahor on (vocal, drums, mix bus, master and so on).
- Which DAW and computer you used.
- Anything that looked wrong, sounded wrong, or was confusing.
- What you loved, and what you wish it did.

Screenshots and short audio examples are especially helpful. Thank you for testing.